Data Types, Variables and Expressions

9.10 Introduction

Every programming language has two fundamental elements, viz., data types and variables. They are very essential elements to write even the most elementary programs. C++ provides a predefined set of data types for handling the data items. Such data types are known as fundamental or built-in data types. Apart from the built-in data types, a programmer can also create his own data types called as User-defined data types. In this chapter, we are going to learn about built-in data types.

9.11 Concept of Data types

Let us look at the following example,

Name = Ram

Age = 15

Average_Mark = 85.6

In the above example, Name, Age, Average_mark are the fields which hold the values such as Ram, 15, and 85.6 respectively.

In a programming language, **fields** are referred as **variables** and the **values** are referred to as **data**. Each data item in the above example looks different. That is, "Ram" is a sequence of alphabets and the other two data items are numbers. The first value is a whole number and the second one is a fractional number. In real-world scenarios, there are lots of different kinds of data we handle in our day-to-day life. The nature or type of the data item varies, for example distance (from your home to school), ticket fare, cost of a pen, marks, temperature, etc.,

In C++ programming, before handling any data, it should be clearly specified to the language compiler, regarding what kind of data it is, with some predefined set of data types.

9.12 C++ Data types

In C++, the data types are classified as three main categories

(1) Fundamental data types

(2) User-defined data types and

(3) Derived data types.

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